KITABU QUEST

S4

Subsidiary Mathematics LFK HLP HGL

Mascot: mountain gorilla
Scene: students using applied mathematics for humanities and language combinations with a mountain gorilla mascot
CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S4 Subsidiary Mathematics LFK HLP HGL

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S4 learners study Subsidiary Mathematics LFK HLP HGL one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Learn through examples, activities, practice, and reflection.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Functions Relations And Graphs
2. Matrices And Transformations
3. Linear Programming And Inequalities
4. Trigonometry And Bearing Problems
5. Mensuration And Similarity Review
6. Statistics And Probability For Decisions
7. Commercial Mathematics And Rates
8. Form IV Examination Practice Project

As you finish each topic, return to the success criteria and correct one answer before moving on.

Functions Relations And Graphs: Lesson Opener

Learning focus: Functions Relations And Graphs

Country link: a Rwandan school, market, farming, building, transport, or data problem for Functions Relations And Graphs.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about functions relations and graphs

Inquiry questions:
- How can functions relations and graphs help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Functions Relations And Graphs: Learn

Functions Relations And Graphs teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Functions Relations And Graphs
- Functions
- Relations
- Graphs
- method
- model
- unit
- answer

Functions Relations And Graphs: Worked Example

Worked example for Functions Relations And Graphs: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Functions Relations And Graphs: Vocabulary And Study Skill

Use these words and study moves before you practise Functions Relations And Graphs. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Functions Relations And Graphs
- Functions
- Relations
- Graphs
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Functions Relations And Graphs without a clear example, method, or evidence.

Functions Relations And Graphs: Guided Activity

Guided activity for Functions Relations And Graphs:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Functions Relations And Graphs: Practice And Correction

Practice for Functions Relations And Graphs:

Main outcome: solve and explain problems about functions relations and graphs.

1. Solve one functions relations and graphs question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to functions relations and graphs and solve it.

Functions Relations And Graphs: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Functions Relations And Graphs.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Functions Relations And Graphs: Outcome Check

Outcome check for Functions Relations And Graphs: Solve and explain problems about functions relations and graphs.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices And Transformations: Lesson Opener

Learning focus: Matrices And Transformations

Country link: a Rwandan school, market, farming, building, transport, or data problem for Matrices And Transformations.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about matrices and transformations

Inquiry questions:
- How can matrices and transformations help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices And Transformations: Learn

Matrices And Transformations teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Matrices And Transformations
- Matrices
- Transformations
- method
- model
- unit
- answer
- check

Matrices And Transformations: Worked Example

Worked example for Matrices And Transformations: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Matrices And Transformations: Vocabulary And Study Skill

Use these words and study moves before you practise Matrices And Transformations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Matrices And Transformations
- Matrices
- Transformations
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matrices And Transformations without a clear example, method, or evidence.

Matrices And Transformations: Guided Activity

Guided activity for Matrices And Transformations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Matrices And Transformations: Practice And Correction

Practice for Matrices And Transformations:

Main outcome: solve and explain problems about matrices and transformations.

1. Solve one matrices and transformations question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to matrices and transformations and solve it.

Matrices And Transformations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matrices And Transformations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices And Transformations: Outcome Check

Outcome check for Matrices And Transformations: Solve and explain problems about matrices and transformations.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Programming And Inequalities: Lesson Opener

Learning focus: Linear Programming And Inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Linear Programming And Inequalities.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about linear programming and inequalities

Inquiry questions:
- How can linear programming and inequalities help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Programming And Inequalities: Learn

Linear Programming And Inequalities teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Linear Programming And Inequalities
- Linear
- Programming
- Inequalities
- method
- model
- unit
- answer

Linear Programming And Inequalities: Worked Example

Worked example for Linear Programming And Inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear Programming And Inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Linear Programming And Inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Linear Programming And Inequalities
- Linear
- Programming
- Inequalities
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear Programming And Inequalities without a clear example, method, or evidence.

Linear Programming And Inequalities: Guided Activity

Guided activity for Linear Programming And Inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear Programming And Inequalities: Practice And Correction

Practice for Linear Programming And Inequalities:

Main outcome: solve and explain problems about linear programming and inequalities.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to linear programming and inequalities and solve it.

Linear Programming And Inequalities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear Programming And Inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Linear Programming And Inequalities: Outcome Check

Outcome check for Linear Programming And Inequalities: Solve and explain problems about linear programming and inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry And Bearing Problems: Lesson Opener

Learning focus: Trigonometry And Bearing Problems

Country link: a Rwandan school, market, farming, building, transport, or data problem for Trigonometry And Bearing Problems.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about trigonometry and bearing problems

Inquiry questions:
- How can trigonometry and bearing problems help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry And Bearing Problems: Learn

Trigonometry And Bearing Problems teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Trigonometry And Bearing Problems
- Trigonometry
- Bearing
- Problems
- method
- model
- unit
- answer

Trigonometry And Bearing Problems: Worked Example

Worked example for Trigonometry And Bearing Problems: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Trigonometry And Bearing Problems: Vocabulary And Study Skill

Use these words and study moves before you practise Trigonometry And Bearing Problems. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trigonometry And Bearing Problems
- Trigonometry
- Bearing
- Problems
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trigonometry And Bearing Problems without a clear example, method, or evidence.

Trigonometry And Bearing Problems: Guided Activity

Guided activity for Trigonometry And Bearing Problems:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Trigonometry And Bearing Problems: Practice And Correction

Practice for Trigonometry And Bearing Problems:

Main outcome: solve and explain problems about trigonometry and bearing problems.

1. Solve one trigonometry and bearing problems question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to trigonometry and bearing problems and solve it.

Trigonometry And Bearing Problems: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trigonometry And Bearing Problems.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Trigonometry And Bearing Problems: Outcome Check

Outcome check for Trigonometry And Bearing Problems: Solve and explain problems about trigonometry and bearing problems.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Mensuration And Similarity Review: Lesson Opener

Learning focus: Mensuration And Similarity Review

Country link: a Rwandan school, market, farming, building, transport, or data problem for Mensuration And Similarity Review.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about mensuration and similarity review

Inquiry questions:
- How can mensuration and similarity review help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Mensuration And Similarity Review: Learn

Mensuration And Similarity Review teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Mensuration And Similarity Review
- Mensuration
- Similarity
- Review
- method
- model
- unit
- answer

Mensuration And Similarity Review: Worked Example

Worked example for Mensuration And Similarity Review: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Mensuration And Similarity Review: Vocabulary And Study Skill

Use these words and study moves before you practise Mensuration And Similarity Review. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Mensuration And Similarity Review
- Mensuration
- Similarity
- Review
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Mensuration And Similarity Review without a clear example, method, or evidence.

Mensuration And Similarity Review: Guided Activity

Guided activity for Mensuration And Similarity Review:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Mensuration And Similarity Review: Practice And Correction

Practice for Mensuration And Similarity Review:

Main outcome: solve and explain problems about mensuration and similarity review.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to mensuration and similarity review and solve it.

Mensuration And Similarity Review: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Mensuration And Similarity Review.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Mensuration And Similarity Review: Outcome Check

Outcome check for Mensuration And Similarity Review: Solve and explain problems about mensuration and similarity review.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Statistics And Probability For Decisions: Lesson Opener

Learning focus: Statistics And Probability For Decisions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Statistics And Probability For Decisions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about statistics and probability for decisions

Inquiry questions:
- How can statistics and probability for decisions help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Statistics And Probability For Decisions: Learn

Statistics And Probability For Decisions teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Statistics And Probability For Decisions
- Statistics
- Probability
- Decisions
- method
- model
- unit
- answer

Statistics And Probability For Decisions: Worked Example

Worked example for Statistics And Probability For Decisions: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Statistics And Probability For Decisions: Vocabulary And Study Skill

Use these words and study moves before you practise Statistics And Probability For Decisions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Statistics And Probability For Decisions
- Statistics
- Probability
- Decisions
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Statistics And Probability For Decisions without a clear example, method, or evidence.

Statistics And Probability For Decisions: Guided Activity

Guided activity for Statistics And Probability For Decisions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Statistics And Probability For Decisions: Practice And Correction

Practice for Statistics And Probability For Decisions:

Main outcome: solve and explain problems about statistics and probability for decisions.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to statistics and probability for decisions and solve it.

Statistics And Probability For Decisions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Statistics And Probability For Decisions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Statistics And Probability For Decisions: Outcome Check

Outcome check for Statistics And Probability For Decisions: Solve and explain problems about statistics and probability for decisions.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Commercial Mathematics And Rates: Lesson Opener

Learning focus: Commercial Mathematics And Rates

Country link: a Rwandan school, market, farming, building, transport, or data problem for Commercial Mathematics And Rates.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about commercial mathematics and rates

Inquiry questions:
- How can commercial mathematics and rates help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Commercial Mathematics And Rates: Learn

Commercial Mathematics And Rates teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Commercial Mathematics And Rates
- Commercial
- Mathematics
- Rates
- method
- model
- unit
- answer

Commercial Mathematics And Rates: Worked Example

Worked example for Commercial Mathematics And Rates: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Commercial Mathematics And Rates: Vocabulary And Study Skill

Use these words and study moves before you practise Commercial Mathematics And Rates. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Commercial Mathematics And Rates
- Commercial
- Mathematics
- Rates
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Commercial Mathematics And Rates without a clear example, method, or evidence.

Commercial Mathematics And Rates: Guided Activity

Guided activity for Commercial Mathematics And Rates:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Commercial Mathematics And Rates: Practice And Correction

Practice for Commercial Mathematics And Rates:

Main outcome: solve and explain problems about commercial mathematics and rates.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to commercial mathematics and rates and solve it.

Commercial Mathematics And Rates: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Commercial Mathematics And Rates.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Commercial Mathematics And Rates: Outcome Check

Outcome check for Commercial Mathematics And Rates: Solve and explain problems about commercial mathematics and rates.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Form IV Examination Practice Project: Lesson Opener

Learning focus: Form IV Examination Practice Project

Country link: a Rwandan school, market, farming, building, transport, or data problem for Form IV Examination Practice Project.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- solve and explain problems about form iv examination practice project

Inquiry questions:
- How can form iv examination practice project help you solve a real problem?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Form IV Examination Practice Project: Learn

Form IV Examination Practice Project teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Form IV Examination Practice Project
- Form
- Examination
- Practice
- Project
- method
- model
- unit

Form IV Examination Practice Project: Worked Example

Worked example for Form IV Examination Practice Project: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Form IV Examination Practice Project: Vocabulary And Study Skill

Use these words and study moves before you practise Form IV Examination Practice Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Form IV Examination Practice Project
- Form
- Examination
- Practice
- Project
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Form IV Examination Practice Project without a clear example, method, or evidence.

Form IV Examination Practice Project: Guided Activity

Guided activity for Form IV Examination Practice Project:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Form IV Examination Practice Project: Practice And Correction

Practice for Form IV Examination Practice Project:

Main outcome: solve and explain problems about form iv examination practice project.

1. Solve one form iv examination practice project question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to form iv examination practice project and solve it.

Form IV Examination Practice Project: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Form IV Examination Practice Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Form IV Examination Practice Project: Outcome Check

Outcome check for Form IV Examination Practice Project: Solve and explain problems about form iv examination practice project.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Functions Relations And Graphs: Review Clinic 1

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Functions Relations And Graphs that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Matrices And Transformations: Review Clinic 2

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Matrices And Transformations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Linear Programming And Inequalities: Review Clinic 3

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Linear Programming And Inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Trigonometry And Bearing Problems: Review Clinic 4

Use this review clinic to strengthen Subsidiary Mathematics LFK HLP HGL without copying earlier answers.

1. Choose one idea from Trigonometry And Bearing Problems that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Functions Relations And Graphs: Application Workshop 1

Application workshop 1 for Functions Relations And Graphs.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Matrices And Transformations: Application Workshop 2

Application workshop 2 for Matrices And Transformations.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Linear Programming And Inequalities: Application Workshop 3

Application workshop 3 for Linear Programming And Inequalities.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Trigonometry And Bearing Problems: Application Workshop 4

Application workshop 4 for Trigonometry And Bearing Problems.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Mensuration And Similarity Review: Application Workshop 5

Application workshop 5 for Mensuration And Similarity Review.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Statistics And Probability For Decisions: Application Workshop 6

Application workshop 6 for Statistics And Probability For Decisions.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Commercial Mathematics And Rates: Application Workshop 7

Application workshop 7 for Commercial Mathematics And Rates.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Form IV Examination Practice Project: Application Workshop 8

Application workshop 8 for Form IV Examination Practice Project.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Functions Relations And Graphs: Application Workshop 9

Application workshop 9 for Functions Relations And Graphs.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Matrices And Transformations: Application Workshop 10

Application workshop 10 for Matrices And Transformations.

Grade move: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

Use this page to turn the skill into a complete problem, not a single short answer.

1. Create one problem from Rwanda school, home, shop, garden, transport, sport, or timetable life.
2. Write the known information and what must be found.
3. Choose a model: table, number line, diagram, equation, graph, shape, or place-value chart.
4. Solve step by step and keep units or labels clear.
5. Check the answer using estimation, inverse operation, substitution, or a second method.
6. Write one sentence explaining why the answer makes sense.

Extension: change one number in the problem and solve the new version.

Glossary

Use this glossary to revise important words in Subsidiary Mathematics LFK HLP HGL.

- Functions Relations And Graphs: Explain this term using an example from the book, your school, or your community.
- Matrices And Transformations: Explain this term using an example from the book, your school, or your community.
- Linear Programming And Inequalities: Explain this term using an example from the book, your school, or your community.
- Trigonometry And Bearing Problems: Explain this term using an example from the book, your school, or your community.
- Mensuration And Similarity Review: Explain this term using an example from the book, your school, or your community.
- Statistics And Probability For Decisions: Explain this term using an example from the book, your school, or your community.
- Commercial Mathematics And Rates: Explain this term using an example from the book, your school, or your community.
- Form IV Examination Practice Project: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Subsidiary Mathematics LFK HLP HGL answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Subsidiary Mathematics LFK HLP HGL that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.